humetrix

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Ms. Stephanie Carlton
Chief of Staff
Centers for Medicare and Medicaid Services
7500 Security Boulevard
Baltimore, MD 21244

Dr. Tom Keane
Assistant Secretary for Technology Policy/National Coordinator
U.S. Department of Health and Human Services
200 Independence Ave SW
Washington, DC 20201

Re: Request for Information; Health Technology Ecosystem (CMS-0042-NC)

As a digital health pioneer that has supported the government's efforts to facilitate patient access to their health data since 2011, Humetrix would like to thank the Centers of Medicare and Medicaid Services (CMS) and the Office of the Assistant Secretary for Technology Policy/Office of the National Coordinator of Health IT (ASTP/ONC) for this important Request for Information (RFI). We are confident that this Administration can break down remaining barriers to full patient access and use of their health data as envisioned in the Health Information Technology Economic and Clinical Health (HITECH) and 21st Century Cures Acts.

Humetrix has a 10+ year history of partnering with the government to provide patients with coverage through Medicare and the VA with an actionable mobile personal health record (PHR).[1] Humetrix won three ONC Industry Innovation competitions for Blue Button with its iBlueButton app (see App store video here: Walkthrough of iBlueButton) – a mobile tool through which Medicare claims-generated PHR data are processed and securely stored on the user's phone for them to view and share as needed with health care providers. Other key milestones in our partnership with the government include:

- In 2018, iBlueButton was the first Medicare approved mobile PHR to access Blue Button 2.0 data. It remains one of the most downloaded Blue Button PHR apps.
- In 2023, Humetrix released the CMS Blue Button 2.0 connected WhatMeds app (see Google Play store video here: WhatMeds Video) for all Medicare beneficiaries with Part D coverage to provide them with an up-to-date list of medications, accompanied

[1] See testimony before the House Energy & Commerce Committee, Subcommittee on Commerce, Manufacturing and Trade here: https://docs.house.gov/meetings/IF/IF17/20160713/105197/HHRG-114-IF17-Wstate-ExpertonB-20160713.pdf. See also testimony before the House Small Business Committee, Subcommittee on Health and Technology here: https://smallbusiness.house.gov/uploadedfiles/6-27-2013_burrow_testimony.pdf.

by personalized patient safety notifications, to help prevent polypharmacy related patient safety issues (responsible for one-third of preventable hospital readmissions in the Medicare population).

- In 2020, Humetrix contracted with the Department of Defense (DoD) Joint Artificial Intelligence Center (JAIC) to provide a dashboard to monitor the pandemic's impact on individuals at higher risk for severe Covid. Humetrix monitored over 18 months 20 million Medicare beneficiaries, receiving and analyzing in real-time 100 million claim lines every week, and by assembling a 7TB database used to build a predictive model to determine at the zip code level areas at highest risk for Covid-19 related hospitalizations to help the military dispatch personnel and supplies when and where needed.[2]

Since 2017, Humetrix has supported the CMS Blue Button 2.0 team in its program development efforts to first launch and then to update the BB2.0 API. **We have repeatedly advocated that CMS launch a large-scale communications effort to tell Medicare patients and their caregivers about the availability of Blue Button connected apps to provide them with their up-to-date medical history for them to share with their provider at every point of care.** We continue to believe that **a top priority for this Administration should be communicating to Medicare patients about the tools they *currently have access to*.**

Additionally, we note that while we are supportive of expanding the Data at the Point of Care (DPC) pilot to general production, we do not believe that this obviates the need for continued focus and effort on Blue Button 2.0. Even with DPC for health systems and other providers, there continues to be immense and complimentary value in providing Blue Button connected apps to Medicare beneficiaries to help them coordinate their own care and ensure patient safety.

Please find below our specific comments, focusing on RFI questions for patients, caregivers, providers, or app developers related to patient data access.

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[2] See the following publications describing Humetrix's work under this DoD JAIC contract: Mapped Medicare Data Helps US Military Deliver Pandemic Support – ARCNEWS – Fall 2021); A Predictive Model for Severe COVID-19 in the Medicare Population: A Tool for Prioritizing Primary and Booster COVID-19 Vaccination – Biology – November 15, 2021; Enhanced Vaccine Effectiveness during the Delta Phase of the COVID-19 Pandemic in the Medicare Population Supports a Multilayered Prevention Approach – Biology - November 24, 2022; and How to Leverage the Medicare Program for a COVID-19 Vaccination Campaign – JAMA Network – November 19, 2020

B. Patients and Caregivers

1. **Patient Needs**

PC-1. What health management or care navigation apps would help you understand and manage your (or your loved ones) health needs, as well as the actions you should take?

a) ***What are the top things you would like to be able to do for you or your loved ones' health that can be enabled by digital health products?***

We have provided consumer support via phone and email for the last 14 years to thousands of iBlueButton users, Veterans, Medicare beneficiaries, and their caregivers. Our app users repeatedly reiterate the immense value of the app, which offers immediate accessibility to critical health information that can be communicated across their health care providers. This includes detailed information on diagnoses, medications, immunizations, procedures, a listing of lab tests and imaging services performed, outpatient encounters, hospitalizations, and health care provider contact information, which no other app provides. For example, iBlueButton and WhatMeds apps' display all dispensed medications with personalized warnings of severe drug-drug interactions (DDIs) based on a ONC high severity drug interaction reference list, opioid risk warnings, and American Geriatrics Society (AGS) Beers list warnings about drugs which are usually inappropriate for patients over 65.

Our users value the ability to review Medicare claims and encounters rendered in a well-organized, consumer-friendly format with the specific provider, encounter or procedure information to identify possible billing errors or fraudulent claims**. Medicare connected apps such as iBlueButton could provide CMS a scalable crowd-sourced means of Medicare fraud detection.**

Reflecting this feedback, Humetrix once again reiterates that it is absolutely critical that CMS communicate to Medicare beneficiaries and their family caregivers about the existence of Blue Button apps and how they can be used to manage health and health care decision-making. Dr. Bettina Experton, CEO of Humetrix, first urged Congress to work with HHS and CMS to undertake a Blue Button communications and education campaign in testimony to the House Energy & Commerce Health Subcommittee in July 2018. Then-U.S. Representatives Susan Brooks (R-IN) and Yvette Clark (D-NY) introduced bipartisan legislation, *H.R. 1390, The Mobile Health Record Act,*[3] to require CMS to address the need for communication to Medicare beneficiaries about the benefits of Blue Button connected apps.

[3] See bill text: https://www.congress.gov/bill/116th-congress/house-bill/1390#:~:text=This%20bill%20requires%20the%20Centers,%2C%20services%2C%20and%20research%20programs

Executive action continues to be necessary to ensure that Medicare beneficiaries know how to access and use their own information. But CMS should not stop there. **To promote wider acceptance and use of Blue Button connected apps for improved care coordination, addressing preventable medical errors causing, and reducing waste and abuse, CMS should not only promote the Blue Button program, but also consider paying for Blue Button connected applications for Medicare beneficiaries to use at no cost to them.** These digital tools can also be especially useful for patients who cross over multiple federal health programs such as Medicare, Medicaid, TRICARE, and the VA Health System.

b) ***If you had a personal assistant to support your health needs, what are the top things you would ask them to help with? In your response, please consider tasks that could be supported or facilitated by software solutions in the future***.

Blue Button connected apps, like the iBlueButton app, immediately provide a detailed medical history, a list of dispensed medications, and a list of beneficiary's healthcare providers with specialty and contact info, along with personalized notifications and reminders. These applications ARE the digital assistants Medicare beneficiaries need.

PC-2. Do you have easy access to your own and all your loved ones' health information in one location (for example, in a single patient portal or another software system)?

a) ***If so, what are some examples of benefits it has provided?***

Medicare Blue Button connected apps that access Medicare Part A, B, and D data and transform these claim data into actionable clinical information – such as iBlueButton app and about 20 of the 75 Blue Button 2.0 connected apps – do provide easy, all-in-one place personal data access for all Medicare FFS beneficiaries with part D coverage. For most of these FFS Medicare beneficiaries, who receive care from multiple providers, Medicare claims data provides one easy-to-use data source to access and leverage the critical clinical information they need to receive safe and cost-effective care.

b) ***If not, in what contexts or for what workflows would it be most valuable to use one portal or system to access all such health information?***

Beneficiaries with coverage through Medicare Advantage may have access to different provider and/or payer portals over time as a result of changes in the payer network and/or plan switching. As a consequence, these beneficiaries lack access to their longitudinal medical history.

CMS should require MA plans to provide their Part C data in a timely manner (at the same frequency that CMS receives Part A or D FFS claim data) so that CMS can make these data available through an updated Blue Button API. Doing so would enable MA plans' enrollees to access their longitudinal health data in one place as FFS beneficiaries are currently able to do. The current payer-to-payer API is intended to ensure a hand off between health plans, but will not serve as a patient-facing resources for health care decision making.

c) ***Were there particular data types, such as x-rays or specific test results, that were unavailable?***

Test results and medical images are not available in claims data. Medicare claims data does contain labs and imaging service dates, as well as the names of facilities, the ordering provider, the reviewing provider, and the diagnoses related to these tests. These data are a valuable proxy to those missing underlying results and images.

PC-3. Are you aware of health management, care navigation, or personal health record apps that would be useful to Medicare beneficiaries and their caregivers?

As mentioned above, about 20 of the 75 Medicare Blue Button 2.0 connected apps, including Humetrix's iBlueButton and WhatMeds, are mobile PHR apps that would be very valuable to Medicare beneficiaries and their caregivers – if they were made more aware of them by Medicare.

PC-4. What features are missing from apps you use or that you are aware of today?

a) ***What apps should exist but do not yet? Why do you believe they do not exist yet?***

Not one, but 20 of these apps exist, and have been approved by CMS to access Blue Button 2.0 data. However, they are listed on the Medicare.gov website in a way that makes them impossible to find. The Medicare.gov website navigation path is exceptionally complex and not intuitive to even those beneficiaries who know what they are looking for.[4]

b) ***What set of workflows do you believe CMS is uniquely positioned to offer?***

No change needs to be made to the Blue Button program, except for making it visible to Medicare beneficiaries and their providers, and for making Part C data available through an updated Medicare Blue Button API.

PC-5. What can CMS and its partners do to encourage patient and caregiver interest in these digital health products?

a) What role, if any, should CMS have in reviewing or approving digital health products on the basis of their efficacy, quality or impact or both on health outcomes (not approving in the sense of a coverage determination)? What criteria should be used if there is a review process? What technology solutions, policy changes, or program design changes can increase patient and caregiver adoption of digital health products (for example, enhancements to data access, reimbursement adjustments, or new beneficiary communications)?

Over the last six years, Humetrix has been asked by the CMS Blue Button team to provide comment and feedback regarding communication and marketing of the Blue Button Apps (now referred to as the "Medicare Connected Apps").

[4] https://www.medicare.gov/providers-services/claims-appeals-complaints/claims/share-your-medicare-claims/connected-apps

Our assessment is that there is a lot of "noise and clutter" on the Medicare.gov website about these Medicare connected apps with many apps or services only tangentially related to the primary purpose of assisting the beneficiaries by leveraging and making the best use of their Blue Button data for managing their healthcare. Some of the "listed apps" only seem to be promoting their existing services for beneficiaries, with no real connection to leveraging the Blue Button data to deliver on the main goals of the Blue Button program

As far as the physical navigation to find and reach the landing page on Medicare.gov for all Medicare connected apps, you have to search very exhaustively on the Medicare.gov website to find the listing of the apps and discover them in the unexpected location about "filing a claim" (as stated above – please see our answer to question PC-4 a).

Here are our specific recommendations for improving communication about Blue Button apps:

- **CMS needs to better categorize and classify the apps it approves in terms of their value and use by Medicare beneficiaries**; specifically, CMS needs to distinguish apps of clinical value (the 20 we referenced above) from those of administrative or marketing nature (e.g., by insurance plans, etc.). The current one-bag fits all category of "organize and share medical information and claims" apps used by Medicare offers a hard-to-understand mixed bag including many apps of minimum value to Medicare beneficiaries.
- **CMS needs to create a Blue Button program communication team with authority to promote the use of Blue Button apps by Medicare beneficiaries** by multiple channels (through the Medicare.gov website, by email, in the annual "Medicare & You" handbook, etc.). The CMS Blue Button 2.0 leadership and their excellent team has on more than one occasion let us know that they do not have the needed authority to control or even much influence on how Medicare communicates about Blue Button apps to Medicare beneficiaries. CMS needs to directly promote the clinical value of its Blue Button program to Medicare beneficiaries to enable them to easily access their health data aggregated in one place and share this information at any point of the care continuum.

b) What changes would enable timely access to high quality CMS and provider generated data on patients?

See our comment above about adding timely Part C data to the Blue Button database.

PC-6. What features are most important to make digital health products accessible and easy to use for Medicare beneficiaries and caregivers, particularly those with limited prior experience using digital tools and services?

Intuitive native mobile apps with elderly-friendly user interfaces should have good contrast and font size to make the information provided understandable and accessible to individuals with visual disabilities. Importantly, the health data content has to be organized for both patients and providers to easily find the information they are looking for, aggregating and deduplicating that data when needed (i.e., repeat prescriptions should be grouped with a fill

history and not repeatedly for every fill, encounters have to be organized and grouped, etc.). Developers should have clinical and healthcare informatics expertise – not only technical expertise to make it right. At Humetrix, all of our applications have been developed by a team of engineers, clinicians, and healthcare informaticists working together and taking user feedback to render our applications as useful and user-friendly as possible.

PC-7. If CMS were to collect real-world data on digital health products' impact on health outcomes and related costs once they are released into the market, what would be the best means of doing so?

If Medicare Blue Button connected apps with clinical utility (the ~20 we mentioned above) were largely promoted and their use tracked by CMS, CMS could conduct prospective outcome analysis of users' vs nonusers of these apps and compare their outcomes, adjusting for needed demographic and clinical beneficiary characteristics.

2. Data Access and Integration

PC-8. In your experience, what health data is readily available and valuable to patients or their caregivers or both?

a) ***What data is valuable, but hard for patients and caregivers, or app developers and other technical vendors, to access for appropriate and valuable use (for example, claims data, clinical data, encounter notes, operative reports, appointment schedules, prices)?***

Over the last 15 years, Humetrix had no issue accessing and making valuable use of Medicare claim data. Note: in 2011, we started by parsing Medicare Blue Button data in text file format! The Medicare Blue Button 2.0 API was of course a big improvement from there, and that API has been very stable and straightforward for us to use.

b) ***What are specific sources, other than claims and clinical data, that would be of highest value, and why?***

To serve the entire Medicare population, we recommend prioritizing the inclusion of Part C data through an updated Medicare Blue Button API.

c) ***What specific opportunities and challenges exist to improve accessibility, interoperability and integration of clinical data from different sources to enable more meaningful clinical research and generation of actionable evidence?***

The under-used, under-recognized opportunity is to promote the use of Medicare claim data though Blue Button connected apps for individual care coordination and population health.

PC-9. Given that the Blue Button 2.0 API only includes basic patient demographic, Medicare coverage, and claims data (Part A, B, D), what additional CMS data sources do developers view as most valuable for inclusion in the API to enable more useful digital products for patients and caretakers?

The use of the word "only" in this question seems to indicate that Medicare claim data are not recognized as the critically valuable "clinical" health data that they are. The issue to be addressed is not the paucity or clinical limitation of Medicare claim data, but the lack of use of Blue Button connected apps because of the lack of communication by Medicare regarding the value and use of Medicare claims data as a source of valuable clinical information for Medicare beneficiairies. As stated above, CMS also needs to include Part C claims data to the Blue Button 2.0 API.

a) ***What difficulties are there in accessing or utilizing these data sources today?***

The only issue is the Part C data missing from the Blue Button 2.0 API.

b) ***What suggestions do you have to improve the Blue Button 2.0 API experience?***

None. Again, there is no technical issue accessing the Medicare Blue Button API which is the most stable and best documented API of all available health data APIs.

c) ***Is there non-CMS data that should be included in the API?***

No – not needed.

C. Providers

1. Digital Health Apps

PR-1. What can CMS and its partners do to encourage providers, including those in rural areas, to leverage approved (see description in PC-5) digital health products for their patients?

a) ***What are the current obstacles?***

The lack of awareness about clinically valuable Medicare Blue Button apps for Medicare beneficiaries or their family caregivers to use and to share with their providers an accurate, organized, and up-to-date medical history, list of meds, prior tests/procedures, etc. Providers should be informed by CMS about these apps and encourage their patients to use them at every encounter. This could be a requirement as part of the annual Medicare Wellness visit and during a Welcome to Medicare visit. A Medicare connected app - of the patient's choice – should be prescribable by providers, and covered by Medicare, as new RPM devices and services are now covered by Medicare through new CPT coding.

b) ***What responsibilities do providers have when recommending use of a digital product by a patient?***

Providers should be able to recommend any of the clinically valuable Medicare approved connected apps (when properly categorized by CMS). Providers' responsibility should be to educate their patients about the importance of always having your up-to-date medical information with them – which these apps can provide.

c) ***Cost or privacy implications of making all data formats interoperable***.

With regards to Medicare connected apps, there is a cost for developers to develop and maintain these apps. This cost – which cannot be borne by patients – should be covered by Medicare.

With regards to privacy, apps should have like the iBlueButton or WhatMeds apps a privacy-by-design architecture, using edge computing and data storage on the user's device only – with no PII or any other data stored in the cloud – to provide optimal individual privacy protection.

E. Technology Vendors, Data Providers, and Networks

1. Ecosystem

TD-1. What short term (in the next 2 years) and longer-term steps can CMS take to stimulate developer interest in building digital health products for Medicare beneficiaries and caregivers?

Getting Medicare approved apps paid for by Medicare when used by Medicare beneficiaries.

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We appreciate the opportunity to provide our feedback on these important questions. Please do not hesitate to reach out if we can be a resource.

Sincerely,

Bettina Experton, MD, MPH
President & CEO
HUMETRIX
1310 Camino del Mar, Suite C
Del Mar, CA 92014
Tel: (858) 259-8987, ext. 210
Cell: (619) 980-5888